

# Covariance for Two Variables

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- Covariance between two variables  $X_1$  and  $X_2$

$$\sigma_{12} = E[(X_1 - \mu_1)(X_2 - \mu_2)] = E[X_1 X_2] - \mu_1 \mu_2 = E[X_1 X_2] - E[X_1]E[X_2]$$

where  $\mu_1 = E[X_1]$  is the respective mean or **expected value** of  $X_1$ ; similarly for  $\mu_2$

- Sample covariance between  $X_1$  and  $X_2$ :  $\hat{\sigma}_{12} = \frac{1}{n} \sum_{i=1}^n (x_{i1} - \widehat{\mu}_1)(x_{i2} - \widehat{\mu}_2)$
- Sample covariance is a generalization of the sample variance:

$$\hat{\sigma}_{11} = \frac{1}{n} \sum_{i=1}^n (x_{i1} - \widehat{\mu}_1)(x_{i1} - \widehat{\mu}_1)$$

- **Positive covariance:** If  $\sigma_{12} > 0$
- **Negative covariance:** If  $\sigma_{12} < 0$

# Covariance for Two Variables

- **Independence:** If  $X_1$  and  $X_2$  are independent,  $\sigma_{12} = 0$  but the reverse is not true
  - Some pairs of random variables may have a covariance 0 but are not independent
  - Only under some additional assumptions (e.g., the data follow multivariate normal distributions) does a covariance of 0 imply independence

□ Example:

|       |   |    |    |
|-------|---|----|----|
| $X_1$ | 1 | -1 |    |
| $X_2$ | 0 | 1  | -1 |

$$\sigma_{12} = E[(X_1 - \mu_1)(X_2 - \mu_2)] = E[X_1X_2] - \mu_1\mu_2 = E[X_1X_2] - E[X_1]E[X_2]$$

$$E(X_1)=?$$

$$E(X_2)=?$$

$$E(X_1X_2)=?$$